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Título

name

Master in Computer Engineering

Supervisor:

Doctor *name, category* Professor,
Iscte – Instituto Universitário de Lisboa

Co-Supervisor:

Doctor *name, category* Professor,
instituição

October, 2022



TECHNOLOGY
AND ARCHITECTURE

Department of Information Science and Technology

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university

October, 2022

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Acknowledgment

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Resumo

Escrever aqui o resumo em Português.

PALAVRAS CHAVE: *palavra_1*, *palavra_2*, ...

Abstract

Write here your abstract in English.

KEYWORDS: *word_1*, *word_2*, ...

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List of Acronyms

AI: Artificial Intelligence

C2DTA: Consumer-Controlled Digital Twin Architecture

C2PDT: Consumer-Controlled Personal Digital Twin

CID: Content Identifier

DID: Decentralized Identifier

DIDComm: Decentralized Identifier Communication

DLT: Distributed Ledger Technology

DSR: Design Science Research

DT: Digital Twin

IoT: Internet of Things

IPFS: InterPlanetary File System

M2M: Machine-to-Machine

MQTT: Message Queuing Telemetry Transport

OEM: Original Equipment Manufacturer

PDE: Personal Data Ecosystem

SD: Smart Device

SLR: Systematic Literature Review

SSI: Self-Sovereign Identity

TD: Thing Description

VC: Verifiable Credential

VDR: Verifiable Data Registry

W3C: World Wide Web Consortium

WoT: Web of Things

CHAPTER 1

Introduction

The International Data Corporation (IDC) [1] estimates that by 2025, Internet of Things (IoT) devices—encompassing machines, sensors, and cameras—will collectively generate 79.4 zettabytes of data. Although Smart Devices (SDs) can increase consumer well-being [2], they also present significant challenges. Sensors invade the consumer privacy sphere, capturing data stored in silos that attract cybercriminals [3], while Original Equipment Manufacturers (OEMs) often exploit these data for their own benefit [4], leaving consumers without tangible rewards [5]. The framework governing the collection and use of personal data involving consumers and OEMs to create economic value, known as the Personal Data Ecosystem (PDE) [6], is unbalanced [7]. It allows OEMs to monopolize data-flow profits while consumers bear the system’s externalities, and is therefore associated with a loss of control over personal information [8].

To address this imbalance, industry and academia are exploring blockchain-based IoT solutions for people-centered architectures [9]. Blockchain’s strong cryptographic foundation, immutable and tamper-proof nature, decentralization, and smart-contract support allow for critical mechanisms that, when integrated with user-centered architectures, elevate consumer status in the PDE [10]. By leveraging blockchain Distributed Ledger Technology (DLT), it is possible to establish a network of trustless peers that collectively agree on an immutable, auditable, and cryptographically secure shared version of the truth [11]. This network offers support for decentralized identifiers that link users with their data without the need for third parties [12]. When combined with Personal Data Stores, frameworks that allow users to collect, store, manage, and share their data according to their preferences [13], this identity mechanism, referred to as SSI [14], gives consumers sovereign control over their data. When used within a user-centered framework, this mechanism transforms consumers from passive data subjects to active participants in the PDE, underscoring the critical role of maintaining data rights—access, usage, and sharing—firmly in the hands of consumers [15].

The DT concept originated at NASA and became publicly recognized in 2003 [16]. It refers to a virtual representation or digital counterpart of a physical object, system, or process [17]. Since its introduction, the concept has evolved and matured, reaching a significant growth stage by 2014 [18]. Transforma Insights [19] forecasts a market size of 24.1 billion devices and a reach of \$1.5 trillion, with the consumer sector accounting for 65% of all connections. DT reframes much of the early efforts to develop connected-device solutions, initially based on telemetry and Machine-to-Machine (M2M) communications and later enhanced with IoT technologies [20]. This evolution has produced a

more structured, multifaceted development framework supporting services and Artificial Intelligence (AI) [17]. For instance, a smartwatch with a DT extends beyond basic data collection: it continuously gathers and analyzes detailed data to update and refine its virtual model, creating a comprehensive understanding of the user’s behaviors, activities, and physiological states.

As DT crosses into the consumer realm, it is reasonable to surmise that some OEMs may exploit the concept of product personalization [21] or other features with disproportionately lower impact for consumers to gain further consumer insights with DT-based solutions. Therefore, it is vital to develop mechanisms that enable consumers to control their devices’ DTs [22]. Tesla is a paradigmatic example of the urgent need for such mechanisms: its trillion-dollar valuation [23] is supported by its ability to leverage DTs to train its Full Self-Driving feature [24], [25].

1.1. Problem Statement

The economic fairness challenge in controlling DTs arises from the current cloud-based approach, where OEMs retain *de facto* control over the DT and the associated consumer-generated data [26]. As consumers become more aware of the growing value of data, they are likely to demand solutions that restore control and ensure equitable participation.

Furthermore, proponents of edge computing, federated learning, and privacy-preservation methods such as homomorphic encryption emphasize the benefits of storing and processing data at the edge as a mechanism to improve the privacy and security of AI training data within IoT contexts [27], [28], [29], [30], [31]. However, these works often overlook the critical step of ensuring that data are kept at the edge in a self-sovereign manner under consumer control.

This dissertation addresses these challenges by presenting the Consumer-Controlled Digital Twin Architecture (C2DTA). The main objective of C2DTA is to empower consumers in the PDE by shifting the DT from the cloud under OEM control to the edge under consumer control [32]. This shift makes consumers *de facto* controllers—meaning they have actual control in practice [33]—of their SD data by giving them effective control over the DT and its associated data [34].

C2DTA addresses two key challenges:

- (1) **Economic fairness through the control of DTs.** The current cloud-based approach allows OEMs to retain *de facto* control over the DT and associated consumer-generated data. C2DTA restores consumer sovereignty by hosting the DT at the edge.
- (2) **Bridging the edge-computing gap.** C2DTA provides the practical foundation necessary for self-sovereign, edge-based data management, closing the gap between proposals for edge computing, federated learning, and privacy-preserving technologies and their actionable implementation.

1.2. Research Objectives

Building on the problem statement, this dissertation pursues the following research objectives:

- RO1.: Self-sovereign DT ecosystem architecture.** Propose an architecture that enables stakeholders of the digital-twinning SD ecosystem to securely manage DTs and leverage their data. In this architecture, those who own devices control the associated DTs at the edge, and stakeholders transact in a decentralized, self-sovereign, and secure manner.
- RO2.: DT, SSI, and blockchain integration taxonomy.** Propose a structured framework for consistently evaluating conceptual and practical implementations, enabling clear classification, identifying research gaps, and guiding future development in decentralized systems.
- RO3.: Hyperledger Fabric, Indy, and Aries integration.** Propose an architecture that integrates Hyperledger Fabric for managing SDs and datasets with Hyperledger Aries for overseeing stakeholder identifiers, Verifiable Credentials (VCs), and secure communication protocols through ACA-Py. This integration encompasses eight detailed use cases across the full SD lifecycle—manufacturing, sale, ownership transfer, twinning, and untwinning.
- RO4.: Future innovation for consumer empowerment.** Introduce concepts emerging from the architecture, including “data-only DTs” and the aggregation of several consumer-controlled DTs into the Consumer-Controlled Personal Digital Twin (Consumer-Controlled Personal Digital Twin (C2PDT)).

1.3. Research Methodology

This dissertation adopts Design Science Research (DSR) as its overarching research methodology. DSR is a problem-solving paradigm rooted in engineering and the sciences of the artificial [35] that seeks to extend the boundaries of human and organizational capabilities by creating new and innovative artifacts [36].

Following the DSR methodology process model proposed by Peffers et al. [37], the research proceeds through six activities:

- (1) **Problem identification and motivation.** The imbalanced PDE, where OEMs monopolize consumer data generated by SDs, is identified as the core problem (Chapter 1).
- (2) **Objectives of a solution.** The research objectives (RO1–RO4) define what the artifact must achieve (Section 1.2).
- (3) **Design and development.** The C2DTA architecture is designed and implemented as the research artifact, integrating DT, SSI, and blockchain technologies (Chapter ??).
- (4) **Demonstration.** The architecture is demonstrated through a business scenario that tracks the lifecycle of a smartwatch equipped with a heartbeat sensor, from

manufacture to purchase, twinning, and resale, encompassing eight use cases (Chapter ??).

- (5) **Evaluation.** Feasibility testing confirms that C2DTA effectively empowers consumers to manage their SD DTs and associated data. Performance benchmarks assess the impact of the decentralized infrastructure (Chapter ??).
- (6) **Communication.** The results are communicated through this dissertation and the accompanying publication [38].

The artifact type is an *instantiation*, as defined by March and Smith [39]: a working system that demonstrates that constructs, models, and methods can be implemented, enabling concrete assessment of the artifact’s suitability to its intended purpose.

A problem-centered approach [37] is adopted, as the research originates from the observation of the imbalanced PDE and the need to empower consumers with control over their SDs’ DTs.

To ground the state of the art, a Systematic Literature Review (SLR) is conducted following the PRISMA guidelines [40] and the methodology proposed by Kitchenham and Charters [41]. The SLR methodology, detailed in Chapter 2, systematically identifies, evaluates, and synthesizes existing research on the integration of DTs, SSI, and blockchain.

1.4. Contributions

This dissertation makes four main contributions:

- (1) **Self-sovereign DT ecosystem architecture.** An architecture that enables stakeholders to securely manage DTs and leverage their data. Device owners control the associated DTs at the edge, and stakeholders transact in a decentralized, self-sovereign, and secure manner. The architecture further explores its impact on the PDE and the business strategies to support it.
- (2) **DT, SSI, and blockchain integration taxonomy.** A structured framework for consistently evaluating conceptual and practical implementations of DT–SSI–blockchain systems. The taxonomy classifies systems along three dimensions: DT implementation level (simplified vs. comprehensive), SSI implementation level (partial vs. full), and blockchain ecosystem implementation level (absent vs. partial vs. integrated).
- (3) **Hyperledger Fabric, Indy, and Aries with ACA-Py integration.** An architecture that integrates Hyperledger Fabric for managing SDs and datasets with Hyperledger Aries for stakeholder identifiers, VCs, and secure communication protocols. The integration encompasses eight use cases across 88 steps, including automation mechanisms to safeguard ecosystem integrity.
- (4) **Future innovation for consumer empowerment.** The introduction of “data-only DTs” and the aggregation of several consumer-controlled DTs into the C2PDT, positioning consumer-controlled Personal Digital Twins as pivotal in empowering individuals in the AI era.

1.5. Document Structure

This dissertation is organized as follows:

Chapter 2 — State of the Art.: Presents the SLR on the convergence of DT, SSI, and blockchain. Covers DT concepts, SSI standards, blockchain for IoT, the PDE, and data sovereignty. Analyzes related work and identifies research gaps.

Chapter ?? — C2DTA Architecture.: Introduces the C2DTA design, detailing the integrated technologies—Eclipse Ditto, Hyperledger Fabric, Hyperledger Indy, ACA-Py, InterPlanetary File System (IPFS), Message Queuing Telemetry Transport (MQTT), and Decentralized Identifier Communication (DIDComm)—and the eight use cases that cover the full SD lifecycle.

Chapter ?? — Results and Discussion.: Presents the implementation, the smartwatch case study, feasibility testing, and performance benchmarks. Discusses the architecture’s impact on the PDE and business strategies.

Chapter ?? — Conclusions and Future Work.: Summarizes the findings, discusses limitations, and outlines future research directions, including the C2PDT concept and its role in the AI era.

CHAPTER 2

State of the Art

This chapter presents a comprehensive review of the state of the art in the convergence of DT, SSI, and blockchain technologies. Section 2.1 describes the SLR methodology adopted. Sections 2.2–2.5 survey the foundational domains. Section 2.6 analyzes the related work, and Section 2.7 identifies the research gaps that motivate C2DTA.

2.1. Systematic Literature Review Methodology

To provide a comprehensive understanding of the state-of-the-art integration of DT with SSI and blockchain, a systematic approach is adopted for the literature review. The SLR follows the PRISMA guidelines [40] and the methodology proposed by Kitchenham and Charters [41].

2.1.1. Research Question

The guiding research question for the SLR is:

“How are DTs integrated with SSI and blockchain?”

The objective is to identify existing implementations for a comparative analysis with C2DTA, classify them according to a structured taxonomy, and identify research gaps.

2.1.2. Search Strategy

The search is conducted across five major scientific databases: IEEE Xplore, ACM Digital Library, Scopus, Web of Science, and SpringerLink. The following search string is used:

`“Digital Twin” AND Blockchain AND (“self-sovereign” OR “decentralized identity” OR “verifiable credential”)`

Non-English publications and research efforts that, while mentioning the search terms, do not discuss—at a minimum—the integration of DTs and SSI are excluded.

2.1.3. Selection Process

The selection process follows the PRISMA workflow [40], illustrated in Figure 2.1. Records are identified through database searching, screened by title and abstract, assessed for eligibility through full-text review, and finally included in the qualitative synthesis.

2.1.4. Classification Taxonomy

The selected papers are categorized based on a three-dimensional classification taxonomy that evaluates the integration levels of DT, SSI, and blockchain within decentralized systems.

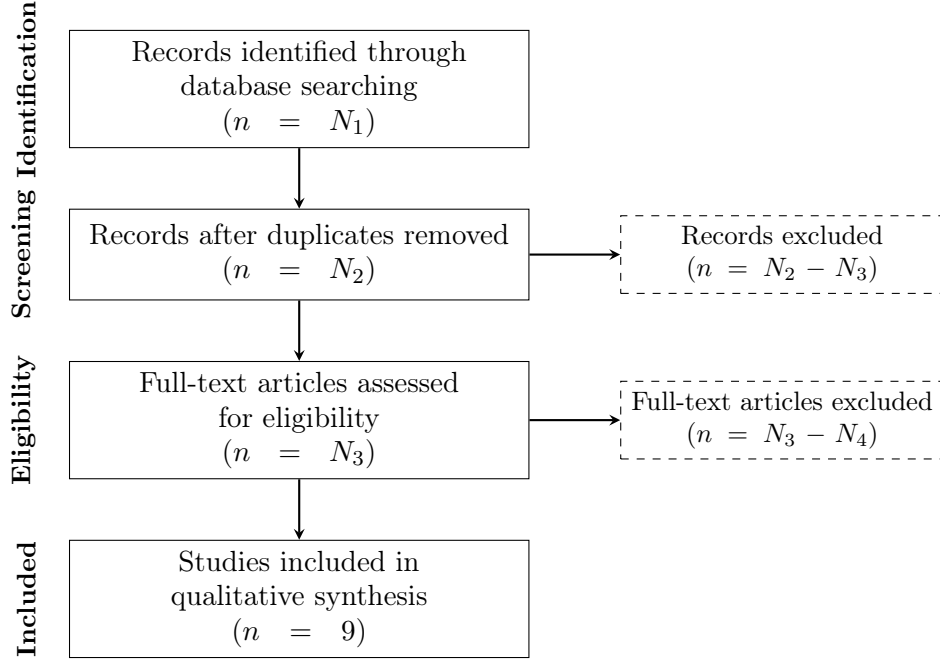


FIGURE 2.1. PRISMA flow diagram for the literature review process.

DT Implementation Level.: Distinguishes between *Simplified* strategies, which represent only specific aspects of an asset or system for niche applications, and *Comprehensive* strategies, which fully represent assets across multiple dimensions via fit-for-purpose platforms capable of IoT integration and real-time updates.

SSI Implementation Level.: Differentiates between *Partial*, which uses only Decentralized Identifiers (DIDs), or DIDs and VCs, and *Full*, which integrates DIDComm protocols for stakeholder communication.

Blockchain Ecosystem Implementation Level.: Evaluates the extent to which blockchain manages ecosystem transactions, with values ranging from *Absent*, where blockchain is not mentioned, to *Partial*, where it is implemented for limited purposes, and *Integrated*, where blockchain fully captures and manages all ecosystem transactions.

Blockchain inclusion emerges naturally with the adoption of SSI, as DIDs and VCs enable a decentralized framework of verifiable and secure exchanges, fostering trust among stakeholders and establishing an ecosystem of inherently trustworthy interactions, recorded and ensured by blockchain.

2.2. Digital Twins

2.2.1. Definition and Evolution

The DT concept refers to a virtual representation or digital counterpart of a physical object, system, or process [17]. The term was first introduced publicly by Grieves in 2003 in the context of product lifecycle management [16]. By 2014, the concept had evolved and matured, reaching a significant growth stage [18]. Glaessgen and Stargel [42]

articulated the DT paradigm for future NASA and U.S. Air Force vehicles, defining it as an integrated multi-physics, multi-scale, probabilistic simulation of a vehicle that uses the best available physical models, sensor updates, and fleet history to mirror the life of its flying twin.

DT reflects the inevitable trend toward cyber-physical integration [17], enabling human interaction to transcend the boundaries of the physical realm, where inherent spatial and temporal constraints can limit efficiency. Ishii and Ullmer [43] introduced the concept of “tangible bits” in 1997, foreshadowing the “bits to atoms” transformation that DT embodies.

Tao et al. [44] propose a five-dimension DT model comprising: (i) the physical entity, (ii) the virtual model, (iii) services, (iv) DT data, and (v) the connections between them. This model provides a comprehensive reference architecture for DT implementations across multiple domains. Fuller et al. [45] conduct a review of enabling technologies, challenges, and open research in DT, confirming the growing interest in the field. Jones et al. [46] provide a systematic literature review characterizing the DT, identifying key themes and future research directions.

2.2.2. Digital Twins in the Consumer Domain

DT reframes much of the early efforts to develop connected-device solutions, initially based on telemetry and M2M communications and later enhanced with IoT technologies [20]. This evolution has produced a more structured, multifaceted development framework supporting services and AI [17]. Transforma Insights [19] forecasts a market size of 24.1 billion devices and a reach of \$1.5 trillion, with the consumer sector accounting for 65% of all connections.

In the consumer domain, a DT extends beyond basic data collection. A smartwatch with a DT, for instance, continuously gathers and analyzes detailed data to update and refine its virtual model, creating a comprehensive understanding of the user’s behaviors, activities, and physiological states. Kopponen et al. [22] propose a blueprint for empowering citizens with DTs. Sahal, Alsamhi, and Brown [47] provide a close look at the Personal Digital Twin in the context of personalized healthcare.

However, as DT crosses into the consumer realm, OEMs may exploit product personalization [21] or other features to gain consumer insights via DT-based solutions, with disproportionately lower benefit for consumers. This underscores the need for mechanisms that enable consumers to control their devices’ DTs [22].

2.2.3. Digital Twin Platforms and the Web of Things

Eclipse Ditto [48] is an open-source DT platform that provides a framework for managing digital representations of physical devices. Ditto implements the World Wide Web Consortium (W3C) Web of Things (WoT) Thing Description (TD) specification [49], which standardizes the description of device capabilities, protocol bindings, and security

mechanisms. The WoT architecture [50] enables interoperability across heterogeneous IoT ecosystems by providing a common vocabulary and interaction model.

Ditto models devices as “Things” with features that represent device capabilities (e.g., sensors, actuators). Each Thing is identified by a namespace-qualified identifier and can be created, read, updated, and deleted through a REST API. Ditto supports multiple protocol bindings, including MQTT, HTTP, and WebSocket, enabling real-time synchronization between the physical device and its virtual counterpart.

Damjanovic-Behrendt and Behrendt [51] discuss an open-source approach to DT design and implementation for smart manufacturing, highlighting the advantages of open platforms. Picone, Mamei, and Zambonelli [52] propose a flexible and modular architecture for edge DTs, demonstrating the feasibility of hosting DTs at the edge with low latency.

2.2.4. Edge Digital Twins

Edge computing brings computation and data storage closer to the sources of data [28]. Satyanarayanan [53] identifies the emergence of edge computing as a fundamental shift in computing architecture, driven by the need for low-latency processing of IoT data.

Hosting DTs at the edge offers several advantages: reduced latency for real-time telemetry, offline resilience when cloud connectivity is intermittent, and enhanced data sovereignty by keeping consumer data under local control. The combination of edge computing with DT technology enables a local-first approach where the DT operates autonomously, synchronizing with external services when connectivity is available.

2.3. Self-Sovereign Identity

2.3.1. Evolution of Digital Identity

SSI emerged as a response to the limitations of user/password internet identity management, which transformed the internet into a “patchwork of identity one-offs” [54] governed by feudal-like contracts of adhesion [55]. Efforts to improve user experience and privacy include the 2001 LinkTank initiative [56], the 2004 Identity Gang [57], and Cameron’s 2005 [54] claims-based architecture.

The concept of sovereignty emerged in 2011 with Loffreto’s [58] proposal for sovereign source authority, arguing for the right to an identity at birth. Around this period, decentralized naming systems such as Namecoin and Blockstack [59], [60] triggered the discussion of blockchain-based identity systems. By 2016, the U.S. Department of Homeland Security had backed the first SSI implementation [14], [61], contributing to the W3C Credentials Community Group and the W3C DID Working Group.

2.3.2. Decentralized Identifiers

DIDs are permanent, globally resolvable, decentralized, and cryptographically verifiable identifiers anchored in a Verifiable Data Registry (VDR) that leverages decentralized ledgers such as blockchain [62]. DIDs rely on a private/public key pair. The identity owner

manages the private key with a digital wallet, while the public key and other metadata are anchored to the blockchain as a JSON structure named DIDDoc.

As DID matured, SSI architects realized that in peer-to-peer interactions, DIDs and DID documents can be generated and exchanged directly between peers needing to identify and authenticate, without registering them on a blockchain. This approach, supported by self-certifying DIDs [14], offers massively better scalability and performance than ledger-based DIDs without compromising security [63].

2.3.3. Verifiable Credentials

VCS are assertions about a DID that can be independently verified via cryptographic methods to ensure their integrity and authenticity [64]. Issuers authenticate VCs with private keys, holders store them in digital wallets, and verifiers use the issuer’s public key to verify their authenticity.

The W3C VC Data Model [65] standardizes the format and exchange of verifiable claims. Three roles are defined: the *issuer*, who creates and signs the credential; the *holder*, who stores and presents it; and the *verifier*, who validates the credential’s authenticity and integrity. This model enables zero-knowledge proofs, where a holder can prove a claim (e.g., age over 18) without revealing the underlying data.

2.3.4. DIDComm

DIDComm enables DIDs and VCs to be exchanged by software agents that represent and act on behalf of an identity owner [66]. Agents hold DIDs, cryptographic keys, and VCs, and interact with other agents via DIDComm [67].

DIDComm is a framework for safe, structured interactions built atop DIDs. It uses a message-based architecture that enables transport-agnostic operation, supporting synchronous, asynchronous, online, and offline scenarios. Together, DIDs, VCs, and DIDComm enable individuals to take control of and manage data about them [68] by fostering an owner-centric, decentralized, standard-based approach to personal data management [69].

2.3.5. Hyperledger Indy and Aries

Hyperledger Indy is a purpose-built distributed ledger for decentralized identity, providing tools, libraries, and reusable components for creating and using independent digital identities rooted on blockchains [14]. Indy uses Plenum Byzantine Fault Tolerant consensus and stores DIDs, schemas, credential definitions, and revocation registries on the ledger.

Hyperledger Aries provides the infrastructure for peer-to-peer interactions based on decentralized identities and VCs. ACA-Py (Aries Cloud Agent – Python) is the reference implementation of a Hyperledger Aries agent [70], providing support for credential issuance, presentation, and verification protocols, as well as DIDComm messaging.

2.3.6. Self-Sovereign Identity for IoT

Fedrecheski et al. [69] provide a perspective on SSI for IoT environments, identifying challenges related to constrained devices, key management, and the need for lightweight protocols. The application of SSI to IoT devices enables each device to possess its own identity, managed through DIDs and VCs, independent of centralized identity providers.

This approach enables secure device authentication, authorization, and data provenance without relying on manufacturer-controlled infrastructure. However, challenges remain in managing cryptographic keys on resource-constrained devices and in ensuring the scalability of DID resolution for large-scale IoT deployments.

2.4. Blockchain for IoT

2.4.1. Blockchain Fundamentals

Blockchain is a DLT that maintains a continuously growing list of records, called blocks, linked and secured using cryptographic hashes [11]. Two broad categories of blockchain exist: *permissionless* (public) blockchains, where any node can participate in consensus (e.g., Bitcoin [71], Ethereum), and *permissioned* (private or consortium) blockchains, where participation is restricted to authorized entities.

Permissioned blockchains are particularly suited for enterprise and IoT applications, as they offer higher throughput, lower latency, and fine-grained access control compared to permissionless alternatives [72]. Smart contracts—self-executing programs stored on the blockchain—enable the automation of business logic and the enforcement of agreed-upon rules without intermediaries.

2.4.2. Hyperledger Fabric

Hyperledger Fabric [72] is a permissioned blockchain platform designed for enterprise use. Its modular architecture separates transaction execution from ordering and validation, enabling high throughput and flexible endorsement policies.

Key architectural components include:

- **Peers:** maintain the ledger and execute chaincode (smart contracts).
- **Orderers:** establish consensus on transaction order using protocols such as Raft.
- **Channels:** provide data isolation between different groups of participants.
- **Chaincode:** application-specific smart contracts written in Go, Java, or JavaScript.
- **Certificate Authorities:** manage identities and issue enrollment certificates.

Fabric’s execute-order-validate architecture enables confidential transactions and supports rich queries against the state database (CouchDB or LevelDB). Guggenberger et al. [73] provide an in-depth investigation of Fabric’s performance characteristics, demonstrating its suitability for enterprise workloads. Allenbrand [74] discusses smart-contract-enabled consortium blockchains for supply chain information management.

2.4.3. InterPlanetary File System

The IPFS [75] is a peer-to-peer distributed file system that uses content-addressed storage. Each file is identified by a unique Content Identifier (CID) derived from its cryptographic hash, ensuring data integrity and immutability.

In IoT contexts, IPFS enables decentralized storage of device data without relying on centralized cloud providers. By anchoring CIDs on a blockchain, it is possible to create an immutable audit trail that proves a given dataset existed at a specific point in time, without storing the full dataset on-chain. Sangeeta and Nam [76] discuss blockchain and IPFS-based data storage for vehicular networks, demonstrating the feasibility of this approach for mobile IoT applications.

2.5. Personal Data Ecosystem and Data Sovereignty

2.5.1. The Personal Data Ecosystem

The PDE refers to the framework governing the collection and use of personal data involving consumers and organizations to create economic value [6]. The current PDE is unbalanced [7]: OEMs monopolize data-flow profits while consumers bear the system’s externalities and experience a loss of control over personal information [8].

The MyData initiative [77] proposes a human-centered model for personal data management, where individuals have the right to know what data are collected about them, the ability to control how those data are used, and the capacity to benefit from the value their data create.

2.5.2. Data Property Rights and Regulation

The question of data property rights is complex and contested. Purtova [78] examines property rights in personal data, analyzing the American discourse. Custers and Margieri [79] argue that the EU fundamental right to data protection is at odds with trade in personal data. Janecek [80] explores ownership of personal data in the IoT, highlighting the tensions between different legal frameworks.

The European Data Governance Act (DGA) [81] relies on neutral data intermediaries to promote data sharing by matching supply and demand [82]. These intermediaries include data marketplaces, platforms, trusts, and personal data intermediaries designed to facilitate effective data exchange and collaboration.

2.5.3. International Data Spaces and GAIA-X

The International Data Spaces (IDS) Reference Architecture Model [83], [84] provides a framework that allows organizations to define terms and conditions related to data sovereignty, such as data usage, pricing, and validity periods. The IDS RAM is at the core of the GAIA-X initiative [85], which aims to develop a federated data and service infrastructure for Europe.

C2DTA champions a distinct viewpoint. Rather than moving data from “cloud silos” to even larger “data space silos,” it proposes anchoring personal data at the edge, where

a decentralized computing grid can leverage federated learning and privacy-preserving technologies such as homomorphic encryption [86] to train AI models. With this approach, C2DTA enables a highly private AI infrastructure where AI models, rather than consumer data, are exchanged. This approach significantly simplifies the privacy and legal challenges associated with data property rights [33], [87], [88], the classification of personal data [80], and the complexities of consent [89], [90].

2.6. Related Work

This section presents the analysis of the nine eligible publications identified through the SLR, classified according to the taxonomy introduced in Section 2.1.4. Table 2.1 summarizes the comparative analysis.

Table 2.1: Literature review results: comparative analysis of DT–SSI–blockchain integration.

Reference	Study Type	Domain	DT Level	SSI Level	Blockchain Level
Cocco & Tonelli [91]	Conceptual	BIM	Simplified	Partial	Partial
Wen & Lin [92]	Implem.	Digital Identity	Simplified	Partial	Integrated
Daniel et al. [93]	Conceptual	General	Comprehensive	Partial	Partial
Thomas et al. [94]	Implem.	Mobility	Simplified	Partial	Integrated
Kiss et al. [95]	Conceptual	Cloud-Edge	Comprehensive	Partial	Absent
Ghirmai et al. [96]	Conceptual	Metaverse	Simplified	Partial	Absent
Ante, Fischer & Strehle [97]	Review	Generic	Simplified	Partial	Partial
Cali et al. [98]	Conceptual	Metaverse / Industry 4.0	Comprehensive	Partial	Absent
Burgos & Pustisek [99]	Conceptual	Industry 4.0	Comprehensive	Partial	Integrated

2.6.1. Analysis of Related Work

Cocco and Tonelli [91] propose a conceptual framework for digital transformation in the construction sector using blockchain, Building Information Modeling, and SSI. Their DT implementation is simplified, acting primarily as a holder of notarized data without dynamic behavior or real-time updates. SSI integration is partial, discussing DIDs and VCs as trust mechanisms but without implementing DIDComm protocols.

Wen and Lin [92] present Twin3, a system for pluralistic personal digital twins via blockchain. The DT is represented by Soulbound Tokens that tokenize user attributes, a simplified approach without IoT integration. While blockchain is fully integrated (ERC-4671 standard), SSI integration is partial, incorporating DIDs and VCs without DIDComm.

Daniel et al. [93] discuss theoretical applications of integrating DT, IoT, AI, and blockchain. While the DT discussion is comprehensive in scope, the work remains conceptual, and SSI is discussed without concrete implementation details.

Thomas et al. [94] propose a novel approach to vehicle sales using DIDs. The DT is represented as an ERC-721 token holding vehicle information but without telematics or IoT integration. Ethereum blockchain is fully integrated for transaction management and off-chain data storage via IPFS.

Kiss et al. [95] present Swarmchestrator, a decentralized framework for orchestrating applications in the cloud-to-edge continuum. DTs are discussed comprehensively as representing computing resources with real-time updates, but blockchain is absent from ecosystem management and SSI is only partially addressed.

Ghirmai et al. [96] explore SSI for trust and interoperability in the metaverse. The DT is discussed solely for holding identity data without dynamic behavior, and both blockchain ecosystem management and DIDComm protocols are absent.

Ante, Fischer, and Strehle [97] provide a bibliometric review of research on digital identity, mapping key research streams. DTs are discussed without dynamic or operational aspects, and SSI is addressed at a general level without concrete implementation details.

Cali et al. [98] discuss SSI in the metaverse for Industry 4.0, positioning the DT to manage identity and lifecycle data across industrial systems. While the DT discussion is comprehensive, blockchain is absent from ecosystem transaction management, and DIDComm is not addressed.

Burgos and Pustisek [99] address trust and scalability of blockchain-based shared manufacturing. DTs are discussed comprehensively for virtualizing physical assets and processes. Blockchain is central to the system, managing trust, transactions, and smart contracts. However, SSI is partially addressed without DIDComm.

2.6.2. Summary of Findings

The analysis reveals that the current state of the art in DT, SSI, and blockchain integration is dominated by conceptual models and partial implementations. Several patterns emerge:

- **DTs as static data holders.** In many studies, DTs serve as static repositories of information without dynamic interaction, real-time updates, or IoT integration. Only a few works discuss comprehensive DT implementations with fit-for-purpose platforms.
- **Incomplete SSI integration.** All reviewed studies implement SSI only partially, discussing DIDs and VCs without leveraging DIDComm protocols for secure, structured communication. This implies that ecosystem interactions have not been fully considered.
- **Variable blockchain integration.** Blockchain is sometimes absent from ecosystem management or used only for limited purposes such as data notarization. Full lifecycle management and transaction handling across the entire ecosystem remain uncommon.
- **No consumer focus.** The majority of studies target Industry 4.0, Building Information Modeling, the metaverse, or generic scenarios. None focus specifically on empowering consumers with control over their SDs' DTs.

2.7. Research Gaps and Positioning

The analysis of the state of the art reveals the following research gaps:

- Gap 1: No comprehensive DT–SSI–blockchain implementation.:** Existing works offer fragmented or conceptual solutions. No study delivers a fully integrated, operational system that combines comprehensive DT management, full SSI (including DIDComm), and integrated blockchain ecosystem management.
- Gap 2: No consumer-focused architecture.:** The literature is dominated by Industry 4.0, BIM, and metaverse applications. No architecture specifically targets consumer empowerment through control of SD DTs.
- Gap 3: No full DIDComm implementation.:** None of the reviewed studies implement DIDComm protocols for secure communication between ecosystem stakeholders. This limits the practical applicability of SSI in DT ecosystems, as it implies that the complexities of peer-to-peer interactions have not been addressed.
- Gap 4: No edge-based consumer-controlled DT with dual-ledger architecture.:** No existing work proposes hosting DTs at the edge under consumer control, combined with a dual-ledger architecture that separates identity management (Hyperledger Indy) from ecosystem transaction management (Hyperledger Fabric).

C2DTA addresses these gaps by delivering a fully integrated, operational system that directly empowers consumers with full control over their SDs' DTs. The architecture

implements a comprehensive DT using Eclipse Ditto, ensuring real-time, dynamic interaction at the edge. The system achieves full SSI integration through Hyperledger Aries and ACA-Py, supporting DIDComm protocols for secure communication. Hyperledger Fabric is fully integrated to manage the SD lifecycle, while Hyperledger Indy manages decentralized identities. The next chapter presents the C2DTA architecture in detail.

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